MUHAMAD AIMAN FAIZ

London, United Kingdom | (+44) 7760 134831 | aimannfaizz@gmail.com | linkedin.com/in/muhamadaimanfaiz
Portfolio: muhamadaimanfaiz.github.io/Data-Analyst

PROFILE

UCL MEng Mechanical Engineering graduate (2:1) with project experience in financial modelling, experimental data analysis and optimisation, including a 25-year project finance model for a cross-border bridge proposal and an industry-linked project with Cannondale that validated a simulation model against experimental data. Experienced in coordinating contractors, technicians and tenants across 60+ properties while maintaining accurate compliance records. Working knowledge of Excel, Python and MATLAB, and currently learning SQL and Tableau. Seeking a graduate data analyst or business analyst role in London.

EDUCATION

University College London (UCL), London, United Kingdom | Sep 2022 – Sep 2026
MEng Mechanical Engineering, Upper Second-Class Honours (2:1)

- Modules: Financial Management for Engineers, Strategic Performance Management, Project Management, Mathematical Modelling and Analysis II, Advanced Computer Applications
- MARA Young Talent Programme (YTP) Scholar

Kolej MARA Seremban, Malaysia | Jun 2020 – Jun 2022

- Cambridge A-Levels: Mathematics (A*), Physics (A), Chemistry (A*), 15/15 points. Gold Award, Varsity Maths 2021. IELTS 7.5

ANALYTICAL PROJECTS

Singapore–Batam Bridge Financial Model | UCL Group Project | Feb 2026 – Apr 2026

- Built, as part of a 10-person team, a 25-year financial model for a USD 800 million cross-border bridge proposal, forecasting annual cash flows, construction and operating costs, and toll and non-toll revenues.
- Applied Net Present Value, cash-flow and scenario analysis to test robustness under different assumptions, with the base case giving an NPV of +USD 152 million at a 5% discount rate.
- Drafted the team's mixed funding strategy: sponsor equity in an SPV, senior bank debt during construction, Viability Gap Funding, guarantees and Asian Development Bank support, then refinancing into long-term bonds once operating.
- Analysed broader economic and strategic impacts, including infrastructure connectivity, trade potential and regional development, supporting an evidence-based assessment of the proposal.

London Bike-Hire Demand Analysis | Personal SQL and Tableau Project, in progress | Nov 2026 – Present

- Analysing Transport for London Santander Cycles journey records in SQL (SQLite) to identify demand patterns by hour and weekday, the busiest start stations and the most popular routes.
- Cleaning and validating journey data by converting date formats and removing duplicate and out-of-range records.
- Using joins and common table expressions to compare morning departures and arrivals by station, with a Tableau Public dashboard planned.

Combustion Data Analysis: Renewable Fuel Blend | UCL Thermofluids Group Lab | Jan 2024 – Mar 2024

- Analysed experimental data from a Ricardo E6 single-cylinder spark-ignition research engine for reference gasoline and a 20% hexanol / 80% octane renewable fuel blend across ignition timings of 10, 13 and 16 CAD BTDC.
- Performed multivariable data analysis linking NOx and CO2 emissions to peak heat-release rate and crank-angle position, identifying the performance trade-offs of changing operating conditions.
- Validated analytical outputs against experimental results and ideal Otto-cycle predictions, using sensitivity analysis and data filtering to assess uncertainty and reliability.

Mountain Bike Frame Optimisation | MEng Final-Year Project, with Cannondale | Sep 2025 – Jun 2026

- Combined experimental data from 16 strain gauges and dial gauge displacement measurements with Abaqus FEA outputs to validate analytical predictions and identify differences between modelled and observed performance.
- Validated the data acquisition circuit (HX711 and Arduino) against beam theory with under 1% axial and approximately 0.83% bending difference, and applied a signal-to-noise threshold to identify the smallest load giving a reliable reading on each gauge.
- Repeated measurements at each load step and calculated means and standard deviations to assess repeatability, flagging channels affected by baseline drift before the data was used for validation.

- Converted raw HX711 digital counts into strain values and organised repeated trials across three load cases into the final datasets used by the numerical team to validate the model.

Swappable Battery Tray: Data-Driven Design Decisions | UCL Design Project | Sep 2023 – Jan 2024

- Screened 15 candidate materials in Granta EduPack and assessed nine shortlisted options on strength-to-weight ratio, thermal performance, cost, corrosion resistance, recyclability and manufacturability.
- Evaluated the design under a 2.943 kN distributed load using analytical calculations and FEA, then applied topology optimisation and generative design to reduce mass from 198.5 kg to 137 kg (30.1% reduction) at an estimated material cost of £615.13.

Additional projects: MATLAB model of pollutant concentration in the Great Lakes, crane failure investigation built on a review of 14 technical documents (2023), water treatment system for Pigeon Creek, Indiana, sized from flow and retention-time calculations (2025), project management plan for a 10-month research expedition covering WBS, schedule, cost breakdown and risk register (2025).

WORK EXPERIENCE

CNC Property Services, United Kingdom | Property Management Operations Intern | Oct 2025 – Sep 2026

- Maintained compliance and maintenance records covering EICR, EPC, CP12 and fire-door remedial works across 60+ residential properties, supporting accurate tracking of statutory and safety-related requirements.
- Managed a high-season inspection schedule tracker, coordinating appointments and workloads across staff members to maintain visibility and effective coverage during busy periods.
- Coordinated maintenance and remedial works with 5–10 contractors and technicians, arranging access, following up progress and progressing outstanding issues through to resolution.
- Communicated technical defects and work requirements to tenants and non-technical stakeholders, adapting the message to each audience.

University College London (UCL), United Kingdom | Room Monitor Officer | Sep 2025 – Jun 2026

- Conducted daily operational inspections across 5–10 teaching rooms, identifying and escalating technical faults before scheduled sessions, and completing structured fault records within defined timelines.

Makan Cafe, United Kingdom | Customer Service and Operations Associate | Oct 2023 – Dec 2024

- Handled customer enquiries and unexpected situations under time pressure, coordinating with kitchen and floor teams to maintain smooth service.

LEADERSHIP AND ACTIVITIES

KUGEM Volleyball Club, United Kingdom | Founder and President | Oct 2022 – Jun 2026

- Led and coordinated 20+ active members, managing training schedules, tournament logistics and team organisation, and guided the team to GoMammoth Open League 2023 Champions, runner-up at Kent Malaysian Games 2024 and bronze at Warwick SEA Games 2025 and Nottingham Malaysian Games 2025.

High Commission of Malaysia, London | Event Coordinator and Sales Assistant | Aug 2024

- Supported event logistics and sales operations for the International London Malaysia Carnival, a large public event, coordinating activities, responding to attendee enquiries and adapting to changing demand.

Cultural and Traditional Music Society, Malaysia | Founder and President | Jul 2020 – Jul 2021

- Founded and led the society, coordinating two performances for audiences of 450+ while managing an approximately RM3,600 budget, equipment procurement and external professional coaches.

Students' Representative Council, Malaysia | Welfare EXCO | Aug 2020 – Aug 2021

- Represented 380+ students, liaising with college management and other stakeholders to identify concerns and progress practical solutions, and designed an online feedback system to support more efficient issue resolution.

SKILLS

- **Analysis and modelling:** financial modelling, Net Present Value (NPV), scenario and sensitivity analysis, multivariable analysis, data cleaning and filtering
- **Tools:** Microsoft Excel, Python, MATLAB, Microsoft PowerPoint and Word. **Currently learning:** SQL, Tableau
- **Business:** project finance, strategic performance management, stakeholder coordination, technical reporting
- **Engineering software:** Fusion 360, SolidWorks, Abaqus, ANSYS
- **Languages:** English, Malay (native), Mandarin (developing)